

MINISTRY OF EDUCATION AND TRAINING

FPT School of Business & Technology



MASTER THESIS PROPOSAL

Major: Software Engineering

Design and Implementation of a Web-Based Virtual Assistant for Japanese Language Learning Leveraging Multi-Agent Systems and Knowledge Graphs to Personalize Learning Pathways for Vietnamese University Students

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1. Introduction and Background

1.1 Problem Description

In recent years, the relationship between Vietnam and Japan has continued to grow in areas such as education, business, and technology. As a result, many Vietnamese university students are interested in learning Japanese to improve their academic and career opportunities. However, Japanese is considered one of the more difficult languages for Vietnamese learners because of its grammar structure, writing systems, and contextual usage.

Currently, many students in Vietnam still study Japanese through traditional classroom methods or mobile applications. Although these approaches are useful for beginners, they often follow a fixed learning path and cannot adapt well to the individual needs of each learner. Some students may want to focus on conversational communication, while others aim to pass the JLPT examination. However, most existing systems provide the same content and pace for all learners.

In addition, students often face difficulties when studying grammar and kanji. In many cases, learners memorize vocabulary and grammar patterns mechanically but struggle to apply them in real communication situations. They also spend a lot of time searching for explanations from multiple sources because learning materials are scattered and not connected logically.

Recently, Large Language Models (LLMs) such as ChatGPT have shown strong potential in education. They can answer questions quickly and support interactive learning. However, these systems still have several limitations. The generated answers may sometimes be inaccurate or inconsistent, especially for language-learning topics that require structured and reliable explanations. Furthermore, general-purpose chatbots usually do not maintain a personalized learning roadmap for each student.

Because of these issues, there is a need for a more personalized Japanese learning support system that can combine conversational AI with structured learning knowledge. This study proposes a web-based virtual assistant that utilizes a simple multi-agent architecture together with a Knowledge Graph to support Japanese language learning for Vietnamese university students.

1.2 Research Objectives

The main objective of this study is to design and develop an AI-based Japanese learning assistant that can support personalized learning for Vietnamese university students.

The project focuses on the following objectives:

1. To build a Knowledge Graph containing Japanese learning materials such as vocabulary, grammar, kanji, and example sentences for JLPT N5 level.
2. To develop a virtual assistant capable of answering Japanese learning questions using Retrieval-Augmented Generation (RAG).
3. To design a simple multi-agent workflow that separates different system responsibilities such as tutoring, learning planning, and learner assessment.
4. To generate personalized study recommendations based on the learner's level and learning goals.
5. To evaluate the quality of the generated responses and the overall performance of the system.

1.3 Scope of the Project

This project focuses on Vietnamese university students who are learning Japanese at beginner level.

The scope of the system includes:

- Supporting Japanese learning for JLPT N5 level only.
- Providing grammar explanations, vocabulary support, and simple learning recommendations.
- Using Vietnamese as the primary explanation language.
- Developing a web-based prototype for demonstration purposes.

Due to time limitations, the project will not include:

- Speech recognition features.
- Advanced handwriting recognition.
- Full commercial deployment.
- Higher JLPT levels such as N3, N2, or N1.

The research mainly concentrates on backend system design, AI integration, and knowledge organization rather than complex frontend development.

2. Proposed Solution

2.1 Solution Description

The proposed system is a web-based virtual assistant for Japanese language learning. The system combines Large Language Models, Retrieval-Augmented Generation (RAG), and a Knowledge Graph to provide more accurate and personalized learning support.

The workflow of the system is summarized as follows:

1. The learner creates a profile and selects learning goals such as JLPT N5 preparation.
2. The system performs a simple placement assessment to estimate the learner's current level.
3. Based on the assessment results, the planner module generates a recommended learning roadmap.
4. During learning sessions, the tutor module answers questions related to grammar, vocabulary, and sentence structures.
5. The assistant retrieves related learning content from the Knowledge Graph before generating responses in order to improve answer accuracy.

Compared to traditional chatbot systems, the proposed solution focuses more on structured learning support and personalized study guidance.

2.2 Software Architecture

The system adopts a modular architecture consisting of four main components:

1. Knowledge Graph

The Knowledge Graph stores Japanese learning content including:

- Vocabulary
- Grammar points
- Kanji
- Example sentences
- Relationships between lessons

Neo4j will be used to manage graph data and connections between learning concepts.

2. Tutor Module

The Tutor module interacts directly with learners. It uses an LLM combined with RAG to answer questions and explain Japanese grammar in Vietnamese.

3. Planner Module

The Planner module generates simple study recommendations based on:

- Current learner level
- Learning goals
- Available study time

4. Assessment Module

The Assessment module provides short quizzes and tracks learner progress during the study process.

The backend services will be implemented using FastAPI and PostgreSQL.

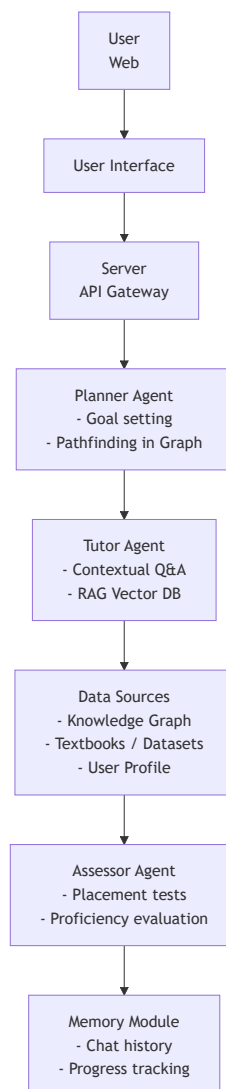


Figure 1: Architectural diagram illustrating the proposed data flow.

2.3 Technology Stack

The project will use several modern technologies to support backend development, AI integration, and knowledge management.

Category	Technology	Purpose
Backend	Python, FastAPI	REST API development
Frontend	Streamlit / Gradio	Simple web interface for prototype
Database	PostgreSQL	Store user profiles and learning progress
Graph Database	Neo4j	Store relationships between learning concepts
AI Framework	LangChain	RAG workflow implementation
Vector Database	Qdrant	Semantic search and retrieval
Development Tools	Git, Postman	Version control and API testing

The selected technologies are widely used, open-source, and suitable for rapid prototype development within the project timeline.

3. Implementation Plan

3.1 Main Development Stages

The project will be implemented in several stages to ensure stable development and testing.

Stage 1: Requirement Analysis and Data Collection

- Study existing Japanese learning applications and AI-based learning systems.
- Collect learning materials for JLPT N5 level.
- Prepare datasets including vocabulary, grammar, kanji, and example sentences.

Stage 2: Knowledge Graph Construction

- Design the structure of the Knowledge Graph.
- Create relationships between learning concepts such as grammar points and vocabulary.
- Store graph data using Neo4j.

Stage 3: Backend Development

- Develop REST APIs using FastAPI.
- Connect PostgreSQL and Neo4j databases.
- Implement user profile and progress tracking functions.

Stage 4: AI Integration

- Integrate Large Language Models through API services.
- Implement Retrieval-Augmented Generation (RAG).
- Develop tutoring and question-answering functions.

Stage 5: Testing and Evaluation

- Test chatbot response quality and retrieval accuracy.
- Measure system response time and database performance.
- Collect feedback from test users if possible.

Stage 6: Final Report and Documentation

- Summarize research findings and implementation results.
- Complete the thesis report and presentation materials.

3.2 Expected Schedule

Week	Activities
Week 1–2	Requirement analysis and dataset collection
Week 3–4	Knowledge Graph design and database setup
Week 5–6	Backend API development
Week 7–8	AI integration and RAG implementation
Week 9–10	System testing and evaluation
Week 11–12	Final report and presentation preparation

3.3 Feasibility Assessment

This project is considered feasible because the research scope is limited to beginner-level Japanese learning (JLPT N5) and focuses mainly on building a working prototype.

Most technologies used in the project are open-source and have extensive documentation and community support. Frameworks such as FastAPI, Neo4j, and LangChain can help reduce development complexity and speed up implementation.

In addition, the required learning materials are publicly available from textbooks and online educational resources. This makes it possible to prepare a structured dataset within the project timeline.

However, some challenges may still occur during development. One important issue is the quality of AI-generated responses. Large Language Models may occasionally produce inaccurate or unclear

explanations. To reduce this problem, the system will combine RAG with structured retrieval from the Knowledge Graph.

Another challenge is time limitation. Because the project duration is relatively short, the system will focus on core functions only, such as tutoring support, retrieval, and personalized recommendations. Advanced features such as speech recognition or mobile deployment are outside the scope of this study.

4. Expected Results

The project is expected to produce the following outcomes:

1. A web-based Japanese learning assistant prototype for Vietnamese university students.
2. A Knowledge Graph containing structured JLPT N5 learning content, including vocabulary, grammar, and kanji relationships.
3. A chatbot system capable of answering Japanese learning questions using Retrieval-Augmented Generation.
4. A simple personalized learning recommendation feature based on learner goals and progress.
5. An evaluation report analyzing the performance and response quality of the proposed system.

The project is also expected to demonstrate how AI technologies and Knowledge Graphs can be applied to support personalized language learning in educational environments.

5. Evaluation Plan

The proposed system will be evaluated using both technical testing and response quality assessment.

The evaluation focuses on the following criteria:

5.1 Response Accuracy

The generated responses will be compared with reference learning materials to determine whether the explanations are correct and relevant to the learner's questions.

5.2 Retrieval Quality

The system will be evaluated on how accurately it retrieves related learning content from the Knowledge Graph and vector database.

5.3 System Performance

Performance testing will measure:

- API response time
- Retrieval latency
- Database query performance
- Stability during continuous usage

5.4 User Feedback

If possible, several students will be invited to test the prototype and provide feedback regarding:

- Ease of use
- Clarity of explanations
- Learning usefulness
- Overall user experience

In addition, the proposed system will be compared with a baseline chatbot system that does not utilize Knowledge Graph retrieval.

6. References

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